

Smart India Hackathon (SIH) Themes

1. MedTech / BioTech / HealthTech

Innovations in healthcare, biotechnology, and health-tech to improve patient care, diagnosis, and treatment.

👉 *Examples:* AI-based disease detection, wearable sensors, hospital management portals, Ayurvedic medicine traceability.

2. Travel & Tourism

Smart solutions to boost India's tourism industry, make travel easy, and promote eco & cultural tourism.

👉 *Examples:* Digital tourism platforms, AR/VR heritage tours, eco-tourism promotion apps.

3. Transportation & Logistics

Tech-driven solutions for urban traffic, logistics, and goods movement.

👉 *Examples:* Smart traffic systems, route optimization, smart public transport apps.

4. Agriculture, FoodTech & Rural Development

Improving farming, rural economy, and food supply chains with technology.

👉 *Examples:* Crop disease detection apps, AI-based fertilizer optimization, direct-to-market farmer platforms.

5. Disaster Management

Smart systems to prevent, manage, and respond to natural or man-made disasters.

👉 *Examples:* Early warning systems, real-time emergency resource allocation, disaster relief planning apps.

6. Smart Education

Tech-based solutions to improve learning, teaching, and student engagement.

👉 *Examples:* AI-driven career guidance, smart timetables, attendance tracking systems, IoT-enabled classrooms.

7. Clean & Green Technology

Environment-friendly technologies for waste management, sanitation, pollution control, and sustainability.

👉 *Examples:* Smart waste segregation bins, pollution tracking apps, eco-friendly packaging.

8. Smart Automation

Using AI and automation to make processes faster, smarter, and more efficient.

👉 *Examples:* Workflow automation, scheduling systems, AI-based decision engines.

9. Blockchain & Cybersecurity

Secure, decentralized, and transparent solutions for data security and digital trust.

👉 *Examples:* Blockchain-based supply chain systems, secure identity verification, cyber threat detection tools.

10. Miscellaneous

Any impactful ideas that don't fit into the above categories but solve real-world problems.

👉 *Examples:* Microplastic detection systems, smart retail apps, sentiment analysis in e-consultations.

11. Renewable / Sustainable Energy

Clean and sustainable energy solutions using solar, wind, or microgrid technologies.

👉 *Examples:* Energy monitoring systems, solar-powered IoT devices, smart grid management.

12. Heritage & Culture

Innovations that preserve and promote India's heritage and culture.

👉 *Examples:* AR/VR cultural experiences, digital heritage preservation, storytelling platforms.

13. Fitness & Sports

Ideas that promote health, fitness, and sports using technology.

👉 *Examples:* Fitness tracking apps, wearables for athletes, sports performance analytics.

S.No.	Problem Statement Title	Category	PS Number	Theme
1	Smart Community Health Monitoring and Early Warning System for Water-Borne Diseases in Rural Northeast India	Software	SIH25001	MedTech / BioTech / HealthTech
2	Smart Tourist Safety Monitoring & Incident Response System using AI, Geo-Fencing, and Blockchain-based Digital ID	Software	SIH25002	Travel & Tourism
3	Low-Cost smart transportation solution for Agri produce from remote farms to nearest motorable road in NER Region	Hardware	SIH25003	Transportation & Logistics
4	Image based breed recognition for cattle and buffaloes of India	Software	SIH25004	Agriculture, FoodTech & Rural Development
5	Image based Animal Type Classification for cattle and buffaloes	Software	SIH25005	Agriculture, FoodTech & Rural Development
6	Development of a Digital Farm Management Portal for Implementing Biosecurity measures in Pig and Poultry Farms	Software	SIH25006	Agriculture, FoodTech & Rural Development

S.No.	Problem Statement Title	Category	PS Number	Theme
7	Development of a Digital Farm Management Portal for Monitoring aximum Residue Limits (MRL) and Antimicrobial Usage (AMU) in Livestock	Software	SIH25007	Agriculture, FoodTech & Rural Development
8	Disaster Preparedness and Response Education System for Schools and Colleges	Software	SIH25008	Disaster Management
9	Gamified Environmental Education Platform for Schools and Colleges	Software	SIH25009	Smart Education
10	Smart Crop Advisory System for Small and Marginal Farmers	Software	SIH25010	Agriculture, FoodTech & Rural Development

S.No.	Problem Statement Title	Category	PS Number	Theme
11	Smart Curriculum Activity & Attendance App	Software	SIH25011	Smart Education
12	Automated Attendance System for Rural Schools	Software	SIH25012	Smart Education
13	Real-Time Public Transport Tracking for Small Cities	Software	SIH25013	Transportation & Logistics
14	Waste Segregation Monitoring System for Urban Local Bodies	Hardware	SIH25014	Clean & Green Technology
15	Intelligent Pesticide Sprinkling System Determined by the Infection Level of a Plant	Hardware	SIH25015	Agriculture, FoodTech & Rural Development
16	Automated Student Attendance Monitoring and Analytics System for Colleges	Software	SIH25016	Smart Education
17	Digital Platform for Centralized Alumni Data Management and Engagement	Software	SIH25017	Smart Education

S.No.	Problem Statement Title	Category	PS Number	Theme
18	Telemedicine Access for Rural Healthcare in Nabha	Software	SIH25018	MedTech / BioTech / HealthTech
19	Digital Learning Platform for Rural School Students in Nabha	Software	SIH25019	Smart Education
20	Development of indigenous contactless Integrated Track Monitoring Systems (ITMS) for Track Recording on Indian Railways	Hardware	SIH25020	Smart Automation

S.No.	Problem Statement Title	Category	PS Number	Theme
21	AI based development of Laser based QR Code marking on 'track fittings on Indian Railways'.	Hardware	SIH25021	Transportation & Logistics
22	Maximizing Section Throughput Using AI-Powered Precise Train Traffic Control	Software	SIH25022	Transportation & Logistics
23	AyurSutra- Panchakarma Patient Management and therapy scheduling Software	Software	SIH25023	MedTech / BioTech / HealthTech
24	Comprehensive Cloud-Based Practice Management & Nutrient Analysis Software for Ayurvedic Dietitians, Tailored for Ayurveda-Focused Diet Plans	Software	SIH25024	MedTech / BioTech / HealthTech
25	E tongue for Dravya identification	Hardware	SIH25025	MedTech / BioTech / HealthTech
26	Develop API code to integrate NAMASTE and or the International Classification of Diseases (ICD-11) via the Traditional Medicine Module 2 (TM2) into existing EMR systems that comply with Electronic	Software	SIH25026	MedTech / BioTech / HealthTech

S.No.	Problem Statement Title	Category	PS Number	Theme
	Health Record (EHR) Standards for India.			
27	Develop a blockchain-based system for botanical traceability of Ayurvedic herbs, including geo-tagging from the point of collection (farmers/wild collectors) to the final Ayurvedic formulation label.	Software	SIH25027	Blockchain & Cybersecurity
28	Smart Classroom & Timetable Scheduler	Software	SIH25028	Smart Education
29	Authenticity Validator for Academia	Software	SIH25029	Smart Education
30	AI-Based Crop Recommendation for Farmers	Software	SIH25030	Agriculture, FoodTech & Rural Development

S.No.	Problem Statement Title	Category	PS Number	Theme
31	Crowdsourced Civic Issue Reporting and Resolution System	Software	SIH25031	Clean & Green Technology
32	Development of a Smart Digital Platform to Promote Eco & Cultural Tourism in Jharkhand	Software	SIH25032	Travel & Tourism
33	AI-Based Smart Allocation Engine for PM Internship Scheme	Software	SIH25033	Smart Automation
34	AI-Based Internship Recommendation Engine for PM Internship Scheme	Software	SIH25034	Smart Education
35	Sentiment analysis of comments received through E-consultation module	Software	SIH25035	Miscellaneous
36	Development of Sensor for Detection Of Microplastics	Hardware	SIH25036	Miscellaneous
37	Development of a low-cost camera-based automated beach sand grain size mapping system	Hardware	SIH25037	Miscellaneous
38	Blockchain-Based Blue Carbon Registry and MRV System	Software	SIH25038	Clean & Green Technology
39	Integrated Platform for Crowdsourced Ocean	Software	SIH25039	Disaster Management

S.No.	Problem Statement Title	Category	PS Number	Theme
	Hazard Reporting and Social Media Analytics			
40	FloatChat - AI-Powered Conversational Interface for ARGO Ocean Data Discovery and Visualization	Software	SIH25040	Miscellaneous

S.No.	Problem Statement Title	Category	PS Number
41	AI-Driven Unified Data Platform for Oceanographic, Fisheries, and Molecular Biodiversity Insights	Software	SIH25041
42	Identifying Taxonomy and Assessing Biodiversity from eDNA Datasets	Software	SIH25042
43	Embedded Intelligent Microscopy System for Identification and Counting of Microscopic Marine Organisms	Hardware	SIH25043
44	AI-Powered Crop Yield Prediction and Optimization	Software	SIH25044
45	Blockchain-Based Supply Chain Transparency for Agricultural Produce	Software	SIH25045
46	Smart Waste Segregation and Recycling System	Hardware	SIH25046
47	Disaster Response Drone for Remote Areas	Hardware	SIH25047
48	Gamified Learning Platform for Rural Education	Software	SIH25048
49	AI-Driven Public Health Chatbot for Disease Awareness	Software	SIH25049

S.No.	Problem Statement Title	Category	PS Number
50	Smart Traffic Management System for Urban Congestion	Software	SIH25050

S.No.	Problem Statement Title	Category	PS Number	Theme
51	Renewable Energy Monitoring System for Microgrids	Hardware	SIH25051	Renewable / Sustainable Energy
52	AR-Based Cultural Heritage Preservation Platform	Software	SIH25052	Heritage & Culture
53	Improved Onion storage technology for enhancing shelf life of onions	Hardware	SIH25053	Agriculture, FoodTech & Rural Development
54	Automated High-Current Short-Circuit Test System for MCB to comply with IEC 60898-1:2015	Hardware	SIH25054	Smart Automation
55	Automated Specimen Preparation System for testing of Cable samples as per IS 10810 and IS 7098.	Hardware	SIH25055	Miscellaneous
56	To explore new/alternative assaying methods to the fire assay method for testing of gold jewellery and artefacts through non-destructive testing	Hardware	SIH25056	Miscellaneous

S.No.	Problem Statement Title	Category	PS Number	Theme
57	Automated Compliance Checker for Legal Metrology Declarations on E-Commerce Platforms	Hardware	SIH25057	Miscellaneous
58	Detection and Prevention of Tampering in Weighing and Measuring instruments	Hardware	SIH25058	Miscellaneous
59	Enhancing Student Awareness on difference between Aadhaar linked and Direct Beneficiary Transfer(DBT) enabled Aadhaar seeded bank account.	Software	SIH25059	Miscellaneous
60	Real life solutions for Waste Management.	Software	SIH25060	Clean & Green Technology

S.No.	Problem Statement Title	Category	PS Number	Theme
61	Digitize and Showcase Monasteries of Sikkim for Tourism and Cultural Preservation	Software	SIH25061	Travel & Tourism
62	Implementation of Smart Agriculture for Efficient Cultivation in Hilly Regions	Hardware	SIH25062	Agriculture, FoodTech & Rural Development
63	Developing a cost effective solution for detecting the breakage of Low Voltage AC Distribution Over Head conductors	Hardware	SIH25063	Disaster Management
64	Improving the Renewable Energy hosting capacity in Distribution Feeders improving the Power quality of Distribution network during high RE injection	Hardware	SIH25064	Renewable / Sustainable Energy
65	Designing and development of an application for on spot assessment of Roof Top Rain water harvesting and artificial recharge potential and size of the RTRWH and AR.	Software	SIH25065	Smart Automation

S.No.	Problem Statement Title	Category	PS Number	Theme
66	Development of an AI-driven ChatBOT for INGRES as a virtual assistant	Software	SIH25066	Smart Automation
67	Proposal for Design and development of application or Heavy Metal Pollution indices	Software	SIH25067	Miscellaneous
68	Real time Groundwater resource evaluation using DWLR data	Software	SIH25068	Miscellaneous
69	AI-Driven Life Cycle Assessment (LCA) Tool for Advancing Circulancy and Sustainability in Metallurgy and Mining	Software	SIH25069	Miscellaneous
70	Secure Data Wiping for Trustworthy IT Asset Recycling	Software	SIH25070	Miscellaneous

S.No.	Problem Statement Title	Category	PS Number	Theme
71	AI-Based Rockfall Prediction and Alert System for Open-Pit Mines	Software	SIH25071	Disaster Management
72	Design and Implementation of Solar-Powered Dewatering in Mining Operations.	Hardware	SIH25072	Renewable / Sustainable Energy
73	AI-Powered Mobile Platform for Democratizing Sports Talent Assessment	Software	SIH25073	Fitness & Sports
74	AI-Powered Personal Farming Assistant for Kerala Farmers	Software	SIH25074	Agriculture, FoodTech & Rural Development
75	Gamified Platform to Promote Sustainable Farming Practices	Software	SIH25075	Agriculture, FoodTech & Rural Development
76	AI-Based Farmer Query Support and Advisory System	Software	SIH25076	Agriculture, FoodTech & Rural Development
77	A hardware that can detect and prevent unauthorized use of electric fences	Hardware	SIH25077	Smart Automation
78	Software other than a circuit breaker that can	Software	SIH25079	Disaster Management

S.No.	Problem Statement Title	Category	PS Number	Theme
	be used to detect and turn off LT lines when the line breaks			
79	Document Overload at Kochi Metro Rail Limited (KMRL)-An automated solution	Software	SIH25080	Smart Automation
80	AI-Driven Train Induction Planning & Scheduling for Kochi Metro Rail Limited (KMRL)	Software	SIH25081	Smart Automation

S.No.	Problem Statement Title	Category	PS Number	Theme
81	Development of a travel related software app that can be installed on mobile phones that could capture trip related information	Software	SIH25082	Travel & Tourism
82	Digital Health Record Management System for migrant workers in Kerala aligned with sustainable development goals.	Software	SIH25083	MedTech / BioTech / HealthTech
83	Earthquake stabilised dialysis system for patient safety during seismic events.	Hardware	SIH25084	Disaster Management
84	Solution for NON-REVENUE LOSS IN WATER SUPPLY Solution to improve AWARENESS IN WATER CONSERVATION TREATMENT OF WASTE WATER AND REUSE FOR DOMESTIC PURPOSES	Hardware	SIH25085	Miscellaneous
85	GREY WATER MANAGEMENT AND REUSE WET LAND MANAGEMENT WATER CONSERVATION	Hardware	SIH25090	Clean & Green Technology
86	AI-Based Timetable Generation System aligned with NEP 2020 for	Software	SIH25091	Smart Automation

S.No.	Problem Statement Title	Category	PS Number	Theme
	Multidisciplinary Education Structures			
87	Development of a Digital Mental Health and Psychological Support System for Students in Higher Education	Software	SIH25092	MedTech / BioTech / HealthTech
88	Centralised Digital Platform for Comprehensive student activity record in HEIs	Software	SIH25093	Smart Education
89	One-Stop Personalized Career & Education Advisor	Software	SIH25094	Smart Education
90	Research and redesign a sport equipment commonly found in sport industry and utilize Fusion software to reimagine its design. Students can use Fusion Features such as Generative Design, Topology Optimization, Additive Build etc. The redesigned component should showcase innovation, enhanced functionality, and improved efficiency, all while being optimized for 3D printing.	Hardware	SIH25095	Smart Education

91	- Students are supposed to use Fusion software to generate NC code with machine details & tool library for any industrial component. - Students should possess technical skills in areas such as CAD/CAM software, G-code programming, toolpath optimization, and machining fundamentals. Additionally, their project ideas should demonstrate a viable solution to a real-world problem, ensuring feasibility and practicality in implementation.	Hardware	SIH25096	Smart Education
92	Research and develop a design on autonomous small precision focused machine for planting crops or weeding.	Software	SIH25097	Smart Education
93	Students are tasked with designing a 4-story commercial office building using Revit Architecture and Revit Structure , ensuring effective integration between architectural and structural models in a BIM environment.(The size of the plot can be assumed by the students, all the dimensions wherever necessary can be assumed by students in mm units.)	Software	SIH25098	Smart Education
94	AI-powered monitoring of crop health, soil condition, and pest risks using multispectral/hyperspectral imaging and sensor data.	Software	SIH25099	Agriculture, FoodTech & Rural Development
95	Accelerating High-Fidelity Road Network Modeling for Indian Traffic Simulations.	Software	SIH25100	Transportation & Logistics
96	Remote classroom for rural colleges	Software	SIH25101	Smart Education
97	AI-based drop-out prediction and counseling system	Software	SIH25102	Smart Automation
98	ERP-based Integrated Student Management system	Software	SIH25103	Smart Automation

99	Language Agnostic Chatbot	Software	SIH25104	Smart Education
100	Develop computer programs (in any language, preferably Python) to identify the design principles behind the Kolam designs and recreate the kolams.	Software	SIH12507	Heritage & Culture

S.No.	Problem Statement Title	Category	PS Number	Theme
101	Development of AI-powered FRA Atlas and WebGIS-based Decision Support System (DSS) for Integrated Monitoring of Forest Rights Act (FRA) Implementation. (States to be concentrated: Madhya Pradesh, Tripura , Odisha, Telangana)	Software	SIH12508	Miscellaneous